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Fundamentals of Machine and Deep Learning using Python

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Part I

Machine Learning



1

Introduction

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You might have seen science fiction movies, and the first thing that comes to mind after hearing the term artificial intelligence (AI) is that some killer robots, some superhuman race with superior thinking and learning capabilities. But AI is far away from all of these. AI is the simulation of human-like intelligence in machines, enabling them to perform tasks such as problem-solving, decision-making, and pattern recognition. Unlike human intelligence, AI lacks emotions, consciousness, and subjective experiences. Human intel-

ligence, on the other hand, encompasses the innate ability to think, reason, learn from experience, and adapt to new situations. It is driven by complex neural networks in the brain and is deeply intertwined with emotions, creativity, and social interactions. While AI excels at processing large volumes of data and repetitive tasks, human intelligence thrives on abstract thinking, moral reasoning, and empathy—qualities that machines cannot replicate authentically. As human beings, we are highly capable of learning new information, also we could work under novel and unseen conditions. However, machines can only work with predefined sequences and in a preprogrammed fashion. **Machine learning (ML) is a subfield of artificial intelligence (AI) that deals with developing algorithms that enable machines to learn from data instead of working on a preprogrammed sequence of operations.** According to Arthur Samuel, one of the pioneers of AI, "ML is a field of AI, which lets machines learn from past data, and improve their performance after a period of time. This allows machines to work on unseen data and in novel situations". AI is a very broad field, driven by the desire to develop systems that possess human-like intelligence, with the power to think and learn like humans. During its earlier stage of evolution, AI gained a pragmatic aspect and computer scientists turned to symbolic AI. Perhaps the earlier success of symbolic AI was a rule-based expert system, which uses a human expert to provide rules for certain domains that were coded into fixed coding logical blocks. One of the most explored aspects of intelligence in computer-based systems is pattern recognition. At its core, ML revolves around pattern recognition, it involves training a computer system to recognize patterns in data and use these patterns to make informed decisions or predictions. This process is based on the analysis of large datasets, which are essential for training ML models effectively.

In the first section 1.2 of this chapter, we will explore the question of what is ML and what it is not. As we said before machines learn from data which means they need a collection of data to get trained and work under novel conditions. In the second section 1.4 we will discuss how to create these datasets, their types, and how to use them for training. The next natural question that should come into our minds is what type of problems an ML algorithm can solve. Is it one algorithm for all or it can solve particular problems, in section 1.5? There are several ways by which ML could be classified one of the possible classifications could be how the ML algorithm can be trained, section 1.6 talks about the classification of ML algorithms. The last section 1.7 discusses ML applications and use cases.

TABLE 1.1

Difference between AI and ML

AI	ML
AI mimics and simulates human intelligence and tries to build machines that have intelligence.	ML is a subfield of AI that deals with learning from past data and solving unseen problems without being explicitly programmed.
The goal of AI is to make a system that can solve either in a human way or by thinking like humans.	ML aims to train some algorithms by applying data for only a limited particular task. And then use that algorithm as a black box by giving it new inputs and generating accurate results.
AI spans some subfields including solving problems by logic, by rules, using ML approaches (learning from data), and others. All the difference is based on how you solve intelligence problems.	ML is a subfield of AI; its subfields include supervised, unsupervised, semisupervised, and reinforcement learning. The aforementioned classification of ML algorithm is based on the choice of ML algorithm training method.

1.1 What is Artificial Intelligence

AI is a field of computer science that deals with the design of algorithms and systems with human intelligence characteristics. So naturally, the first question is how intelligent and in which ways humans are intelligent? Human intelligence encompasses a wide range of cognitive abilities, and AI endeavours to replicate various aspects of it. Some key aspects of human intelligence that AI aims to replicate include:

Reasoning and Problem-Solving Humans excel at logical reasoning and problem-solving, and AI systems, particularly symbolic AI and expert systems, attempt to emulate this by using formal logic and rules to solve complex problems.

Natural Language Understanding and Generation Understanding and generating human language is a complex task that AI models like natu-

ral language processing (NLP) and chatbots strive to replicate. They aim to comprehend and generate text and speech similar to how humans do.

Perception and Sensory Processing AI in the form of computer vision and speech recognition aims to replicate human perception and sensory processing. These systems use neural networks to interpret visual and auditory information, similar to how humans perceive their environment.

Learning and Adaptation Human intelligence is characterized by the ability to learn from experiences and adapt to new situations. Machine learning and deep learning algorithms attempt to replicate this aspect by training on data and improving their performance over time.

Creativity and Innovation Human creativity involves the ability to think beyond existing knowledge and create entirely novel concepts and ideas. While AI systems can generate creative outputs, such as art and music, they often do so by learning patterns from existing data.

Emotional Intelligence AI is making strides in replicating emotional intelligence to some extent, with sentiment analysis and emotion recognition algorithms. However, truly understanding and expressing human emotions remains a significant challenge.

Common Sense Reasoning Human intelligence often relies on common sense knowledge to make everyday decisions. AI systems are working on acquiring and using this kind of knowledge to improve their decision-making abilities.

Social Intelligence AI, particularly in chatbots and social robots, aims to replicate aspects of human social intelligence, such as understanding social cues, showing empathy, and engaging in human interactions.

One of the aspects which humans have and machine shines is pattern recognition. One of the sub-fields of AI is ML which deals with the pattern detection aspect of human intelligence. ML uses a large set of data as an example to discover underlying patterns in the data. Once trained ML algorithms can use this learning to solve unseen problems. The next section 1.2 explains what is ML in the context of AI.

1.2 What is Machine Learning

ML is a subset of AI in which machines learn from data and can work for novel and unseen data. Table 1.1 highlights these differences between AI and ML. At this point, it is worth discussing how ML algorithms learn from data and can work without being explicitly programmed. Figure 1.1 shows this at a very high level, in step 1, the dataset is divided into two sub-parts i.e. training and testing.

In step 2, the training part is used to train the ML algorithm resulting in trained model. The model is ready to be used on unseen data, for this purpose

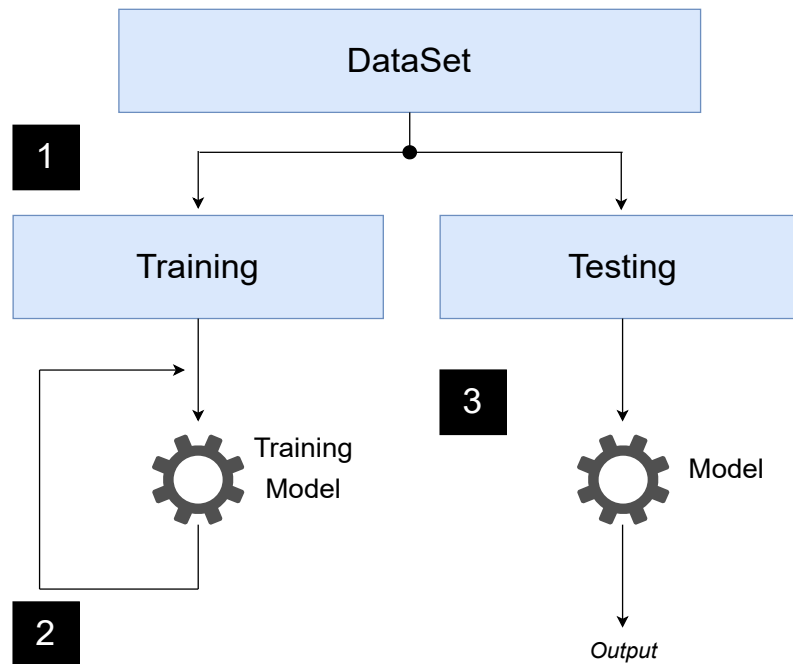


FIGURE 1.1
Machine Learning Cycle

the testing data is passed to the model. In step 3, the model acts as a black box system where input is applied to it resulting in the output. In this way, an ML system can work on novel data without being explicitly programmed.

1.3 An example of explicit programmed versus machine learning

An algorithm is defined as logic that determines how the input would translated into output, or in other words how the machine is going to react to the given set of input. In most cases, the algorithm is explicitly programmed i.e. the programmer writes the program and hard wire it in the machine as shown in 1.2. In many situations, this procedure might not work, for example, in email spam detection if all rules of spam emails are written beforehand, the spammer may catch the algorithm and modify the emails to dodge the system. Therefore, the software should have the ability to dynamically update itself while continuously learning from the data. This approach learns patterns from

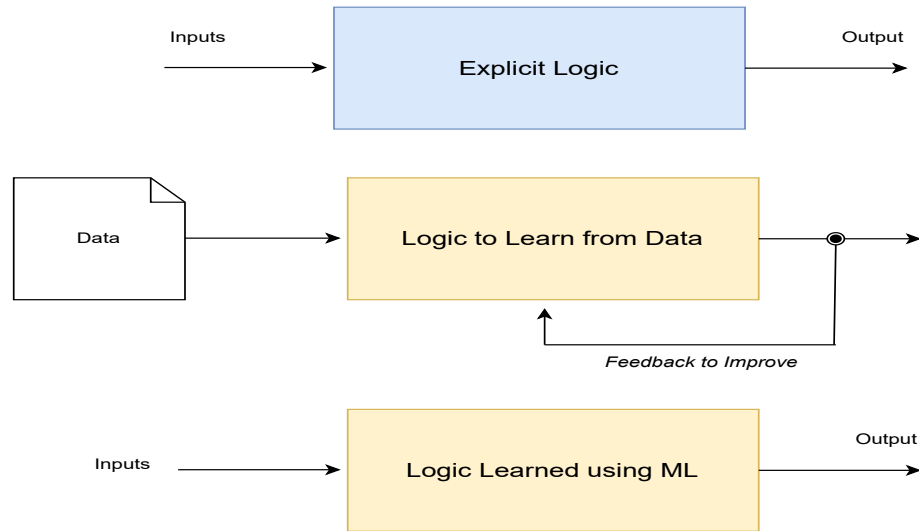


FIGURE 1.2
Explicit versus ML

data and construct an algorithm based on the input data. The major advantage of this approach is the change in the data would trigger an update in the algorithm. Let us consider a fire detection system with two inputs, temperature sensed from the room and smoke which gives True or False signals. The system has a single output which turns the alarm on. In the case of explicit programming, the programmer might write a program defining a rule that governs how to generate the alarm.

```
1 if temp>37 and smoke==True:
2     alarm=True
```

ML does not write the explicit program as above but instead, uses data as shown in table 1.2. In this case, the programmer does not write any explicit logic instead it learns patterns from the data. In this case, the last columns of Fire in table 1.2, will specify two binary labels. The ML algorithm will discover rules from data for classes 0 and 1. The major advantage of this approach is that if data changes the algorithm will adapt dynamically.

1.4 Dataset

Data plays a pivotal role in ML, as most ML algorithms learn using data. The process of learning also known as **training** requires a dataset. The collection of

TABLE 1.2

Dataset for fire detection system

Sr. No.	Temperature	Smoke Sensor	Fire
1.	30	0	0
2.	31	1	0
3.	36	1	1
4.	35	0	0
5.	41	0	1
6.	33	0	0

data either text, images, videos, audio, or other forms is known as **data set**. A dataset has a sample in it, where each sample represents a single entry. In most cases each sample of the data set has a label attached to it, the label defines what the sample is about, it also known as **ground truth**. The dataset in which each sample has an explicit label is called **labeled training dataset**. It is important to note that not all datasets might have labels attached to them, in such case data set would be known as an **unlabeled dataset**.

1.4.1 Dataset

As already discussed that dataset would be different samples $s_i \in S$, where set S has N samples as shown in equation 1.1.

$$S = \{s_1, s_2, \dots s_N\} \quad (1.1)$$

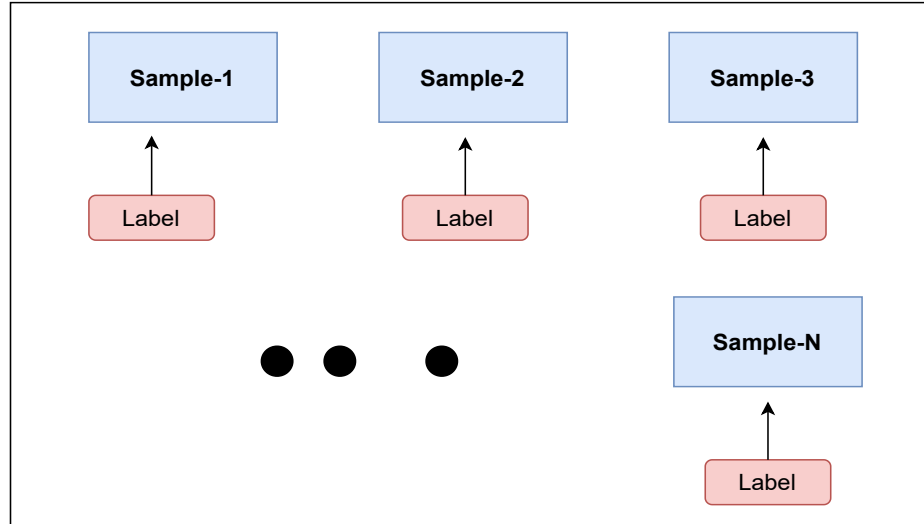
Each sample would have a label $l_j \in L$ attached to it, where set L has K entries as shown in equation 1.2.

$$L = \{l_1, l_2, \dots l_K\} \quad (1.2)$$

The Dataset D of size M creates ordered pairs between each sample s_i and labels l_j as shown in the equation 1.3.

$$D = \{(s_{i1}, l_{j1}), (s_{i2}, l_{j2}), \dots (s_{iM}, l_{jM})\} \quad (1.3)$$

In set D we can count how many samples are there for any label l_j , Let

**FIGURE 1.3**

Labeled training data set

us imagine that we had three labels i.e. $K = 3$. In Set D if the count for l_1 , l_2 , and l_3 is more or less comparable we will say that data set D is **balanced dataset**. In a balanced dataset, the number of samples might not be exactly equal there can be a difference of less than 5%, and if the difference exceeds 5% we will say that the set is imbalanced. In the imbalanced dataset, the labels having more samples are called **majority class**, while the labels that are less represented are called **minority class**. We will discuss this further in section 1.6.1.1.

1.5 Problem in ML

There are three basic problems that we will examine in the context of ML, classification, regression, and clustering. We will employ ML to solve these problems, subsection 1.6 talks about classification in which we will assign a label to each sample of the data set. In subsection 1.5.2 regression we will discuss the second problem in which ML tries to solve the problem. In this case, known data points are plotted in a way that a best-fit line could pass through them. Using this line, the value of the points which were not provided in the initial dataset could be calculated. In subsection 1.5.3, the third problem of ML is presented in which data is divided into groups called clusters based on their common characteristics.

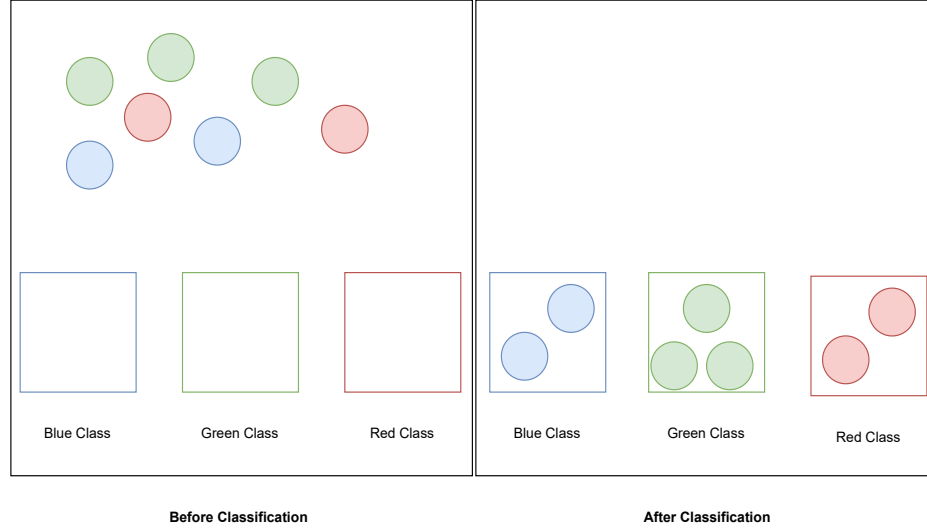


FIGURE 1.4
Classification using colored balls example

1.5.1 Classification

Classification is one of the fundamental problems of ML. The **basic goal of classification is to place each given sample in one of the predefined classes**. To understand classification let us take a very basic example of a heap of colour balls having three types red, green, and blue as shown in figure 1.4. There are three bins also known as classes, each of red, green, and blue colour. The classification task involves placing the ball in the respective colour bin, as shown on the right-hand side of figure 1.4. In ML, the bins are called **classes** while the colour ball is called **samples**. The number of classes is predefined and fixed, and the sample can only be placed in a set of predefined classes. The placement in certain classes is based on certain features, for example, the feature in the ball example was colour. **The feature is anything that could distinguish one sample from the other.** In the example mentioned earlier, there was just a single feature however, there might be more than one feature on which classification can be based. So far we have seen classification using a single feature, however, there could be classification based on multiple features. Let us consider that a teacher wants to classify students in the class based on their height and weight into four classes, TallHeavy (TH), TallLight (TL), ShortHeavy (SH), and ShortLight (SL). The teacher collects the height and weight of all students in the class and tabulates them as shown in the table 1.3.

To divide data into these four classes, the teacher needs to define **bound-**

TABLE 1.3

Data collected from the class students

Sr. No.	Weight	Height
1.	83	5.9
2.	81	5.5
3.	87	5.2
4.	71	5.5
5.	70	5.1
6.	73	5.4
7.	65	5.8
8.	66	5.7
9.	94	5.10

ary of each. Let's imagine that a height greater than 5 feet 7 inches would be considered tall and vice versa. Also, weight greater than 75 kg. would be considered heavy and visa versa. In most cases a variable is defined for each feature, Let's consider H for height and W for weight, equation 1.4 shows the boundary condition for each class. These boundaries are all known as **heuristics**.

$$\begin{aligned}
 W \leq 75, H \leq 5.7 &\Rightarrow SL \\
 W \leq 75, H \geq 5.7 &\Rightarrow TL \\
 W \geq 75, H \leq 5.7 &\Rightarrow SH \\
 W \geq 75, H \geq 5.7 &\Rightarrow TH
 \end{aligned}
 \tag{1.4}$$

1.5.1.1 Binary versus multinomial classification

Classification could be broadly classified into two categories i.e. Binary and multinomial classification. The key difference between them lies in the number of classes or categories that the model is trying to predict. Binary classification is a type of classification problem where the target variable or outcome variable

can take one of two possible classes or categories. Examples of binary classification problems include spam email detection (spam or not spam), disease diagnosis (disease or no disease), and sentiment analysis (positive or negative sentiment). In case of Multinomial classification, also known as multiclass classification, is a type of classification problem where the target variable can have more than two possible classes or categories. Examples of multinomial classification problems include image classification (recognizing different objects or animals), text classification (categorizing news articles into topics like sports, politics, or entertainment), and hand-written digit recognition (classifying digits 0 through 9). In binary classification, the output is often represented as 0 and 1, where 0 corresponds to one class (e.g., "negative" or "no") and 1 corresponds to the other class (e.g., "positive" or "yes"). In multinomial classification, there are more than two possible output values, and each class is typically represented by a unique label or integer.

TABLE 1.4

Mouse population versus pesticide sprayed in area

Sr. No.	Amount of Pesticide spray in area	No. of Mouse in the farm
1.	5500 Liters	4300
2.	5900 Liters	3800
3.	6200 Liters	2200
4.	2500 Liters	5200
5.	8900 Liters	1200
6.	6200 Liters	1700
7.	5400 Liters	2200
8.	4700 Liters	2600
9.	6100 Liters	1800

1.5.2 Regression

Regression models a relationship between two or more variables. To understand regressions let's consider it for the two-variable case. The first variable would be dependent while the other is independent. Regression models the relationship between two variables, in a most simplistic form this relationship would be a close fit straight line. Consider the table 21.1 having mouse population and pesticide sprayed by the farmer. The points of the table are plotted in the table 1.5. It could be observed that the data is not linear and there is not a single straight line that could fit all the points exactly, so regression provides a line that is the best possible fit to all given points.

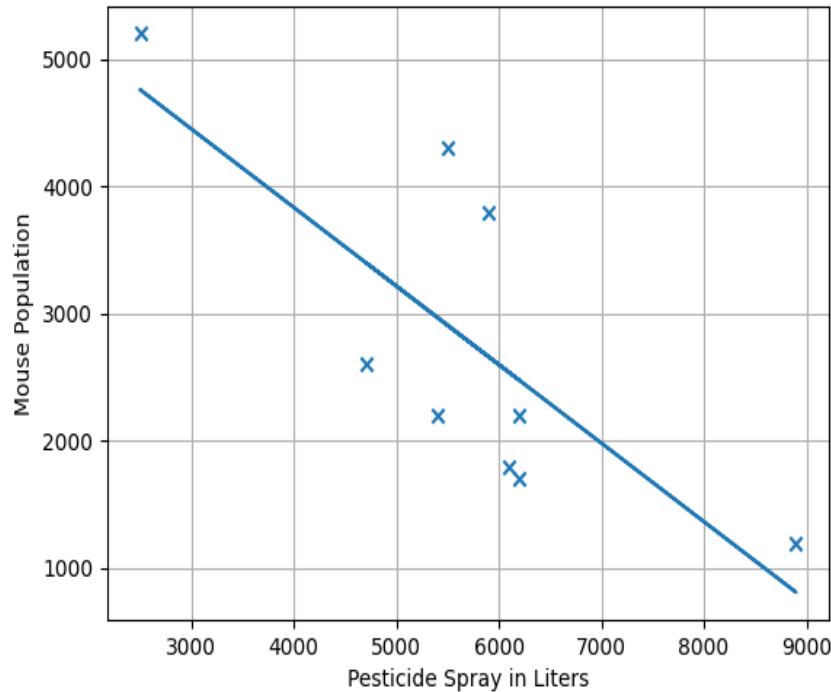


FIGURE 1.5

Regression for farmer spray and mouse population

Once an algorithm is trained on the data, it could give us clues about unseen data. This is possible in two ways, Firstly all the missing data points are now joined by the line, and at any point for any variable other values can be determined. Secondly, the line can be extended on both sides so we can know the points which are actually outside the range of training data points.

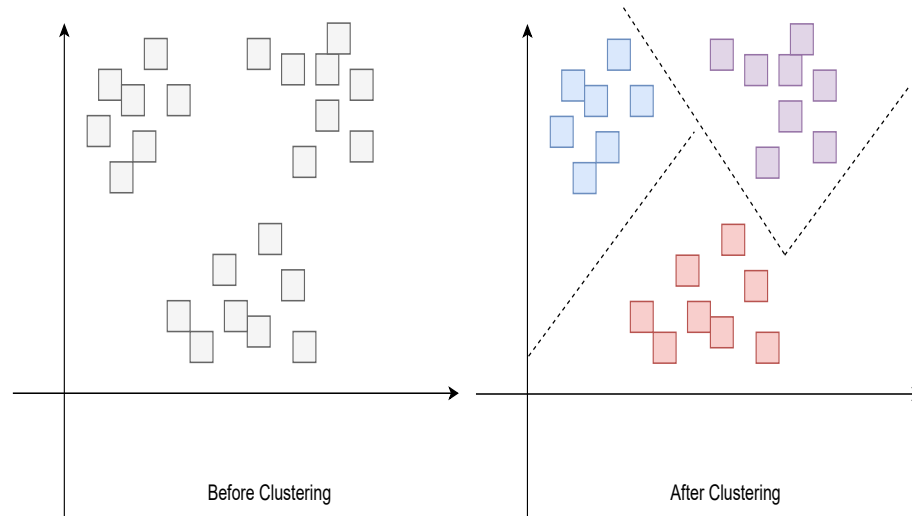


FIGURE 1.6
The Clustering Problem

Find points in regression that are not part of the original dataset is called **prediction** or **forecasting**. Such analysis is useful in several applications such as investment portfolios, predicting sales from customer patterns, determining stock prices, weather predictions, and other areas. The key difference between classification and regression is that for any given query classification returns the class label while regression returns a numerical value.

1.5.3 Clustering

As discussed in section 1.4 not all datasets are labelled, both classification and regression need labelled training datasets. What about an unlabelled training data set? With such data, we can neither train our algorithm for classification problems nor we could determine the relationship between dependent and independent variables for a regression problem. For such data, we could, however, use common characteristics between samples of the dataset to group them into so-called clusters. As discussed in section 1.6 placing each new sample in a certain class would need heuristics which are created on features. In clustering problems, the algorithm would use such underlying features to automatically place each sample in one of the clusters. Placing a sample in a cluster resembles the placement of the sample in each class in a classification problem, however, there is a fundamental difference. In classification, we have explicit heuristics, while in clustering our algorithm would automatically discover the heuristics. Just like classification the number of clusters is predefined and known. As shown on the left side of the figure 1.6, the data does not have

any label attached to it, but it may have some characteristics. The data is divided into three clusters based on their common characteristics on the right side of the figure.

1.6 Classification of ML algorithms

ML algorithms can be classified based on how machine learning algorithms use datasets as explained in sub-section 1.6.1. The second classification can be based on how ML models are trained, in subsection 1.6.2 we presented batch versus incremental learning. The third type of classification is ranked based on how the ML system is generalized. In subsection 1.6.3 we discuss instance and model-based learning.

1.6.1 Based on Datasets

There are different ways in which ML algorithms can be applied depending on the type of problem to be solved and the available dataset. Type of ML could be broadly divided into four categories, firstly Supervised ML subsection 1.6.1.1, in which labelled dataset is available, UnSupervised ML subsection 1.6.1.2 where the dataset is unlabelled and no ground truth is available. Another similar technique is semisupervised learning where the dataset is partially labeled i.e. some samples have labels while others have no labels. Another field Reinforcement ML allows systems to learn from trial and error from feedback obtained by continuous interactions.

1.6.1.1 Supervised ML

Supervised ML is a type of machine learning which works with the labelled training dataset. The machine is trained by a process where each sample of data is given to the algorithm along with the labels. It applies the same way of learning as a student might learn from the teacher. Figure 1.7 shows how labels and data are used to obtain the model after training is completed.

Once the training is completed the model learns how to map any particular input x_i to one of the predefined labels $\hat{y}$. Hence the model is a function $\hat{y} = f(x_i)$ which defines the mapping between input and output based on training data, as shown in figure 1.8.

Every machine learning problem consists of determining a straight line. For example, in a classification with two classes, the line would act as a decision boundary. In the regression problem, the line is a mapping between dependent and independent variables. For elementary mathematics, an equation of a straight line is given in equation 1.5.

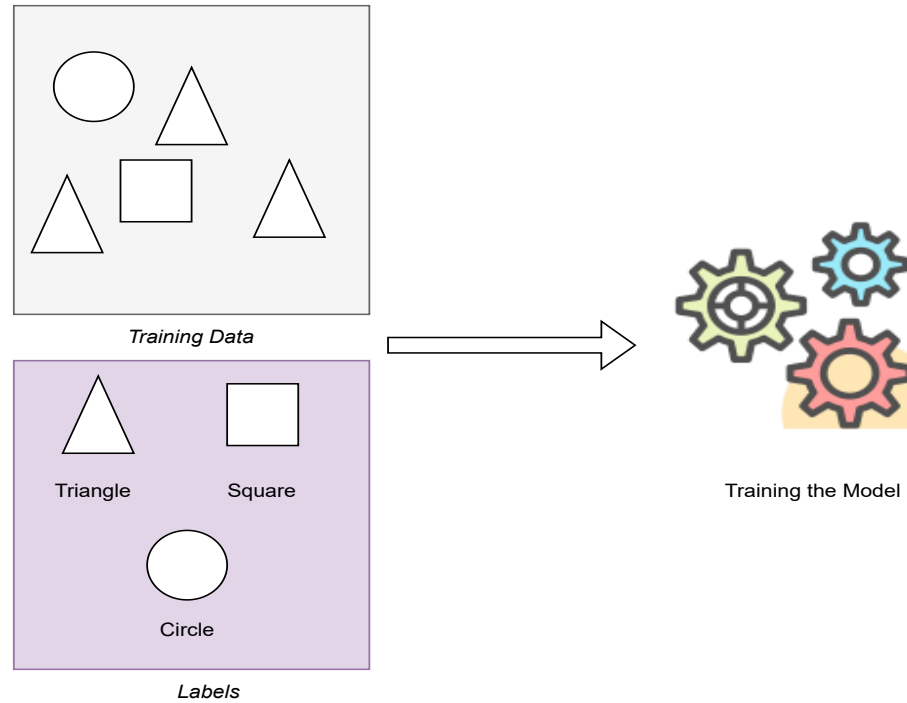


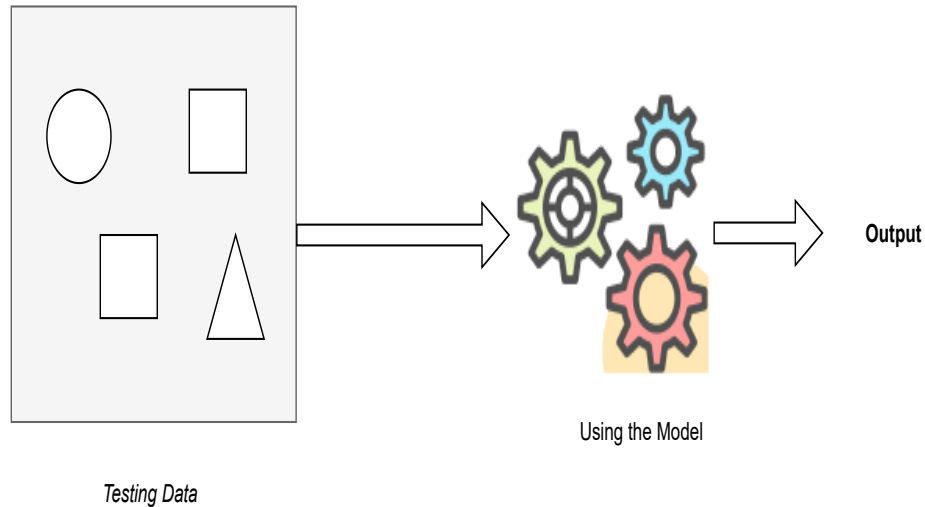
FIGURE 1.7
Training in supervised Learning

$$Y = mX + C \quad (1.5)$$

Where m , is the slope of the line and C is the y-intercept. The equation of a line is an example of a function that maps x to y . The training process of the supervised learning algorithm helps determine the ML algorithm the slope m and y-intercept C . Therefore, a set of data in case of a regression problem helps fit the line to past data, so each new point x_i is projected to regression to find value y . For classification problems, class labels help to learn the decision boundary which is in fact slope and the point where the line cuts y axes. The learned line acts as a heuristic as given in equation 1.4 for any new point.

1.6.1.2 UnSupervised and Semi-Supervised ML

Supervised training algorithms require labelled training datasets, which might not always be possible. Unsupervised learning algorithms work in situations where the data set is unlabelled and discover distribution patterns using a systematic approach. Clustering described in subsection 1.5.3 is a good example

**FIGURE 1.8**

Testing in supervised Learning

of the unsupervised learning algorithm. Since there is no corresponding label to the given data point x_i , the mapping function can only be determined using some statistical attributes such as mean or variance. Unsupervised learning can solve two major problems namely clustering and association without any labels in the data.

- Clustering helps divide data into a predefined number of k groups.
- Association helps to discover and learn rules on any given dataset.

In some cases, the data might be completely labelled and some parts of the data have labels that are useful for learning labels for unlabelled data. Semi-supervised allows algorithms to learn from the part of data that is labelled and use it to group the unlabelled part of data. For example, several phones get group pictures based on the partial labels given by the users. The use might also provide corrections on some parts of data if the grouping is incorrectly performed.

1.6.1.3 Reinforcement ML

Before understanding Reinforcement learning (RL) let's define some key terms:

- Agent is a piece of software that solves a certain problem.
- Environment is everything around the agent.

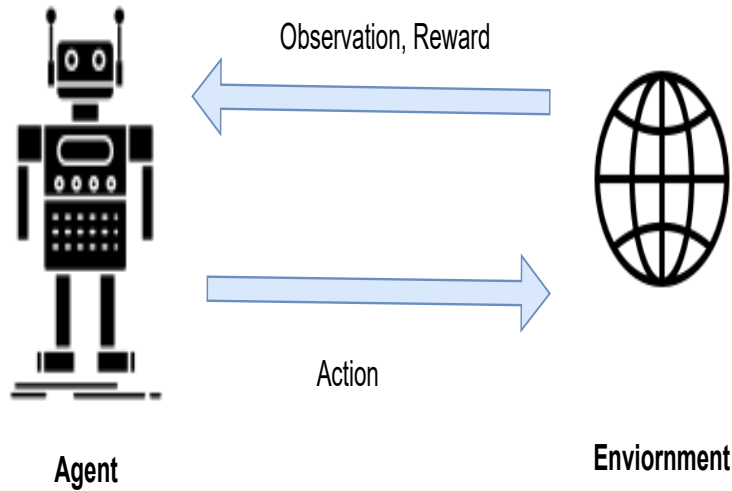


FIGURE 1.9
Reinforcement Learning

- The agent senses from the environment also called observation, and against each observation, it takes a certain action.

Figure 1.9 shows the agent and its interaction with the environment. RL is a type of machine learning in which an agent learns from interacting with the environment. The learning method is based on penalization and reward for every action taken by the agent. There is no particular dataset, labeled or unlabeled, the agent learns from trial and error. For example, when a person learns to ride a bicycle the balancing act comes from the gut feeling and the way the bicycle tends to tilt and fall on one side.

1.6.2 Batch versus incremental learning

ML can be classified into Batch or incremental learning depending on how ML models are trained. Section 1.6.2.1 explains the batch learning while section 1.6.2.2 explains incremental learning

1.6.2.1 Batch learning

In Batch learning model is trained on the entire dataset at once, the entire dataset is loaded into memory, and the model is trained on this dataset in a single batch. The model's parameters (weights and biases) are updated only after processing the entire dataset. Batch learning has two major advantages, firstly, it is straightforward to implement and understand because it processes the entire dataset in one go. Secondly, training on the entire dataset can lead to

more stable and consistent model updates. However, it is not recommended at places where the system is dynamic and the model requires frequent updates. Mostly Batch learning is suitable when the dataset is relatively small and can comfortably fit in memory. Also, when there is no need for online or real-time learning, the model can be retrained periodically with new data.

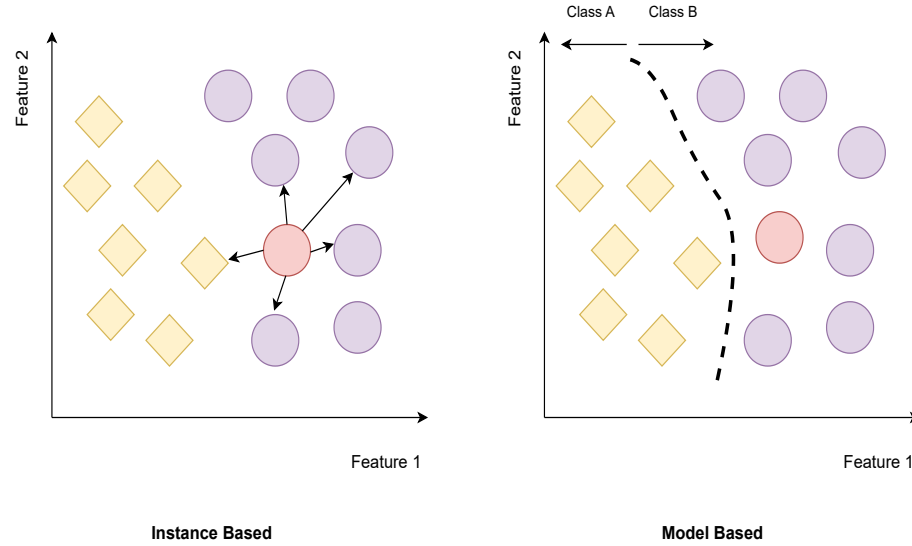
1.6.2.2 Incremental learning

The second type of ML known as incremental learning, also known as online learning or streaming learning, is an approach where the model is updated continuously as new data becomes available. In incremental learning, the model is initially trained on a subset of the data. As new data points arrive, they are used to update the model iteratively. The model's parameters are updated after processing each data point or a small batch of data points. Incremental learning has several advantages, firstly, it is suitable for scenarios where the dataset is too large to fit in memory, as it processes data in smaller chunks. Secondly, incremental learning allows models to adapt to changing data distributions and learn from new data as it arrives. It finds use cases where making predictions or decisions in real-time, models need to adapt to new data continuously. Besides it is suitable in cases where the dataset is too large to load into memory all at once, incremental learning allows for efficient training.

1.6.2.3 Comparison

1. **Efficiency:** Batch learning is generally more computationally efficient as it optimizes the model's parameters over the entire dataset in a single pass. Incremental learning can be slower because it updates the model more frequently, but it's essential for handling large and continuously changing datasets.
2. **Memory Usage:** Batch learning may require more memory since it loads the entire dataset, while incremental learning processes data in smaller chunks, making it more memory-efficient.
3. **Adaptability:** Incremental learning excels when adaptation to new data is critical, while batch learning may require periodic retraining to incorporate new information.
4. **Complexity:** Batch learning is conceptually simpler, while incremental learning requires more intricate algorithms and careful management of model updates.

The choice between batch learning and incremental learning depends on the specific requirements of the problem at hand. Some applications may benefit from a combination of both approaches, using batch learning for periodic updates and incremental learning for real-time adaptation to changing data.

**FIGURE 1.10**

Instance versus model-based learning

1.6.3 Instance versus model-based learning

Instance-based learning and model-based learning are two fundamental paradigms in machine learning, each with distinct approaches and use cases. This note provides an overview of these two learning methods, highlighting their key characteristics, advantages, and applications.

1.6.3.1 Instance based learning

Instance-based learning is an ML approach that relies on storing and memorizing training data instances. Instead of constructing a generalized model, instance-based learning makes predictions based on the similarity between new data instances and the stored training instances. The left side of figure 1.10 shows instance-based learning where a diamond shape represents one class while a circle represents another class. After training there is no generalization that is learned, after a new sample is included in the system the distance with all neighboring points is calculated. As the next step, the number of points near the new point is calculated and averaged to obtain to which class these points belong.

1.6.4 Model-Based Learning

Model-based learning, also known as the model-based method is an ML approach that aims to construct a predictive model from training data. Instead

of directly memorizing instances, model-based learning captures underlying patterns, relationships, or rules in the data to make predictions on new data. The right-hand side of figure 1.10 shows model-based learning where the decision boundary separates Class A and Class B.

1.7 Applications of ML

Machine learning spans a broad range of applications. It shines in every situation where pre-programmed rules are hard to gather or maintain in the long term. However, on the flip side, it needs data for training, so it can only work in situations where data can be collected. In some situations where data is not available reinforcement learning described in section 1.6.1.3 can be applied. Although ML can be applied to a very wide range of applications, this section describes a few applications of ML.

1.7.1 Natural Language Processing

NLP encompasses systems that deal with systems that process text and perform different tasks on it. For example, sentimental analysis of Twitter or Facebook data using text posted by users. In this case, the ML algorithm can tell if the given text has a positive or negative sentiment. Similarly, topic categorization of a given set of documents is another popular application of NLP. In topic categorization, text is analyzed, and based on words that appeared in the text a document is placed in one of the predefined categories. Another well-known NLP application is machine translation. For example, Google Translate uses machine learning to translate text between languages. Chatbots and Virtual Assistants are AI-powered chatbots like Siri, and Alexa, and chatbots for customer service use NLP. The aforementioned are a few applications of ML in NLP, and there are several other applications in which ML can be used.

1.7.2 Recommendation Systems

ML algorithms power recommendation engines like those used by Netflix, Amazon, and Spotify, suggesting content or products based on a user's preferences and behavior. The personalized suggestions to users for products, content, or services are made based on patterns and similarities of users' earlier preferences and choices made by other users with similar interests. In summary, ML algorithms analyze this data to learn patterns and similarities. For instance, if you enjoy action movies, the system may notice that many users who like action movies also enjoy thrillers. Armed with this knowledge, the recommendation system can then suggest items you're likely to enjoy. If you've

been enjoying action films, it might recommend a thrilling action-packed movie you haven't seen yet. The more you use the system, the better it becomes at understanding your tastes. It learns from your feedback, such as the items you rate, like, or ignore.

1.7.3 Financial Services and Marketing

ML is applied to find the right customers and do the marketing to the right audience. For example, it can be used for Customer Segmentation, and find the right audience which have an interest in certain products. Grouping customers based on behavior is very helpful in targeted marketing translating into higher sales. Similarly, ML can be used to predict the likelihood of users clicking on certain advertisements also known as Click-Through Rate (CTR). ML is also widely used in the finance sector, for example, before giving a loan to a customer or a business it can assess the credit risk this is also known as Credit Scoring. It helps save banks from possible defaults and financial losses in the future. Another very popular area is fraud detection in online payments, which particularly uses anomaly detection. An interesting application of ML in finance is the prediction of stocks for investment portfolios.

The presented are a few examples of possible applications of ML, there are several other areas and domains where ML can be applied. In general, ML can be used for tasks involving data analysis, prediction, classification, clustering, optimization, and decision-making across various domains and industries. It can be applied to any problem where patterns or insights can be derived from data, without specifying specific areas or examples.

